

Moosa Alam

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EDUCATION

University of Guelph

Bachelor of Computing, Computer Science (Co-op), Area of Emphasis: Cybersecurity

Guelph, ON

Expected 2029

TECHNICAL SKILLS

Networking: TCP/IP, IPv4/IPv6, Ethernet, Subnetting, VLANs, Routing & Switching, DNS, DHCP, NAT, ACLs, VPNs, SSH

Systems & Tools: Windows 10/11, Linux, PC Hardware, Workstation Setup, System Troubleshooting, Wireshark, tcpdump

Programming: Python, Bash, C, Git

EXPERIENCE

Software Development Intern

PixelsBoost Website Design

Milton, ON

Apr 2026 – Aug 2026

- Built and deployed full-stack client applications serving **1,000+ users**, supporting reliable production systems from requirements through launch.
- Integrated secure Stripe payment infrastructure processing over **\$10,000** in transactions while maintaining reliable application workflows.
- Diagnosed and improved application performance by **32%** through caching, resource optimization, and reduction of network and rendering overhead.

Software Engineering Intern

Vibez Music

Toronto, ON

Jan 2026 – Apr 2026

- Improved internal workflow throughput by **40%+** by developing Python and Flask tools that automated repetitive operational processes.
- Improved system reliability and troubleshooting through structured logging, request validation, error handling, and traceable service responses.
- Automated testing and deployment with GitHub Actions CI/CD while diagnosing and resolving integration and production issues.

PROJECTS

Virtualized Network Home Lab | *Linux, Docker, GitHub Actions, Wireshark*

- Built a segmented multi-host Linux network lab using isolated subnets, configuring IPv4 addressing, routing, DNS, DHCP, NAT, SSH, and firewall rules.
- Deployed containerized services through GitHub Actions CI/CD and validated connectivity using ping, traceroute, and application-level network tests.
- Captured and analyzed traffic with Wireshark and tcpdump to troubleshoot DNS resolution, TCP connections, routing behavior, and packet flow.

User-Space VPN Tunnel | *Python, Linux, TUN/TAP, UDP, Wireshark*

- Built a point-to-point VPN using Linux TUN interfaces and UDP sockets to encapsulate and route IP packets between isolated network endpoints.
- Configured IP forwarding, routing tables, NAT, firewall rules, and authenticated encryption to securely transport network traffic through the tunnel.
- Analyzed packet flow with Wireshark and tcpdump to diagnose connectivity, latency, routing, and packet-loss issues across the tunnel.

CERTIFICATIONS

AWS Certified Cloud Practitioner

AWS

Expected Sep 2026

Cisco Certified Network Associate (CCNA)

Cisco

Expected Dec 2026